Louis Euyseok Han

www.louis-han.info | louishan@engineering.upenn.edu | (267) 456-5667 | Canadian Citizen

Education

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA — May 2027
M.S.E., Mechanical Engineering and Applied Mechanics **(Concentration: Mechatronic and Robotic System)**

Seoul National University, Seoul, Korea — Feb 2017
B.S., Mechanical and Aerospace Engineering

Skills

Programming & Data Engineering: Python, C++, SQL, Shell Scripting, Linux, Test-Driven Development, Docker
Web & Backend Technologies: Django, FastAPI, React, HTML/CSS/JavaScript, AWS, PostgreSQL, Docker
Project Management & Collaboration: Git, GitHub Actions (CI/CD), Atlassian Jira, Confluence, Notion
CAD Tools & Prototyping: SolidWorks, 3D Print(Formlabs), Laser Cutting, Shop Tools, Rapid Prototyping
Mechanical & Electrical: Mechatronics, Feedback Control, Manufacturing, Materials, ROS2, System Integration
Soft Skills: MS Word, Excel, PowerPoint, Problem Solving, Team Leadership, Public Speaking, Patent Prosecution

Experience

University of Pennsylvania, Philadelphia, PA — Oct 2025 - Present
Research Assistant @ Micro Sensors & Micro Actuators Group (Prof Mark G. Allen)

- Developed and experimentally evaluated an aluminum-air battery system for sustainable robotics
- Designed and prototyped flow-cell system for controlled electrolyte circulation to improve Al-air battery performance

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea — Jun 2025 - Aug 2025
Research Intern @ Advanced Machine Intelligence Lab (Prof Tae-hyun Oh)

- Built a pipeline with PyTorch, NumPy, and vision-language models to predict material properties of objects from RGB and NIR images. Integrated the estimated values into 3D point clouds to create property-annotated 3D models

Patsol, Seoul, Korea — Oct 2024 - Jan 2025
Software Developer, Cofounder

- Architected and implemented end-to-end RAG patent-search pipeline using Python, HuggingFace, Elasticsearch, enabling inventors to query patents via natural language instead of inefficient Boolean keyword filters
- Led roadmap presentation to SpringCamp VC and secured $71K seed funding by demonstrating scalable retrieval architecture and subscription-based monetization strategy

Yogiyo (Food Delivery Platform, formerly Delivery Hero Korea), Seoul, Korea — Dec 2022 - Sep 2024
Back-end Software Developer

- Improved portal websites and self-service tools for 500,000+ restaurant owners and optimized internal backoffice tools for Yogiyo staff using RESTful APIs, automated testing, and CI/CD, reducing ~40K monthly manual support tickets
- Automated ingestion of ~1M daily food-delivery orders with Airflow, eliminating manual workflows; collaborated with SRE to scale compute resources for ETL workloads
- Designed pre-aggregated order datasets and optimized database schemas to reduce query latency (from 20s → ~2s). Built scalable Django REST analytics APIs to power order-based sales dashboards

Korean Intellectual Property Office (KIPO), Daejeon, Korea — Jun 2022 - Dec 2022, Apr 2018 - Feb 2019
Patent Examiner

- Conducted prior-art searches and substantive examination for 40+ patent applications per month, evaluating novelty, inventive step, and industrial applicability under Korean patent law
- Drafted numerous Office Actions, including Notices of Reasons for Rejection and Final Rejections

Technical Projects

Wharton AI Hackathon (Hack-AI-thon) (Python, Machine Learning, Prompt Engineering)

- Advanced to the Hackathon finals by developing an AI-powered hotel review question generation system for Expedia using machine learning, generating targeted follow-up questions to elicit more specific and diverse feedback.

Dual Motor Autonomous Car with Web Dashboard (ESP32, C++, PID Control, JS, SolidWorks, ToF Sensors)

- Built an ESP32-based dual-motor robotic car featuring PID speed control, a real-time Wi-Fi web dashboard, and autonomous point-to-point navigation using ToF-based wall detection on a custom SolidWorks-designed chassis

Leadership Experience

UPenn Garage Lab (machine shop and prototyping facility), Lab Instructor — Oct 2025 - Present
UPenn Mechanical Engineering Graduate Association, Vice President — Sep 2025 - Sep 2026
APEC, Intellectual Property Expert Sub-Group Delegate — Aug 2018

- Represented the Republic of Korea in APEC intellectual property meetings as a KIPO Patent Examiner